



ScienceDirect

Journal of Materials Science & Technology

Date: 20 July 2026

[Show more](#) ▼Research article Full text access [Get rights and content](#) ↗

Research Article

Interfacial engineering via thermal corrosion: Achieving dual wettability enhancement and thermal expansion gradient transition in sapphire-SiO₂f/SiO₂ brazed joints

Zhenyu Ye ^a, Yong Xia ^a, Meng Liu ^a, Zijia Wang ^a, Mengting Yang ^a, Jiping Chen ^a, Peixin Li ^a, Jinchun Tu ^b, Jian Cao ^a, Yaotian Yan ^a , Junlei Qi ^a [Show more](#) ▼

View PDF

Download full issue

Cite

Add to Mendeley

Share

10.1016/j.jmst.2025.09.048 ↗

Article

Recommended articles

Cited by (5)

Metrics

Outline

Highlights

Abstract

Graphical abstract

Keywords

1. Introduction

2. Experimental section

Highlights

- This work proposed a novel thermal corrosion-assisted brazing method.
- Na₂SiO₃ generated on the SiO₂f/SiO₂ surface facilitated the wetting reaction.
- A thermal expansion coefficient gradient transition was achieved in the SiO₂f/SiO₂-sapphire joint.
- The shear strength of the brazed joint improved from 21 MPa to 52 MPa.

Abstract

The demand for multi-band optical transmission makes it challenging for a single material to meet the requirements for fabricating radome windows, with the continuous advancement of radar technology. Herein, NaOH thermal corrosion combined with AgCuInTi filler was employed to address the poor wettability of SiO₂f/SiO₂ composites, enabling low-stress, high-strength joining with sapphire. The corrosion products enhanced the surface energy of the SiO₂f/SiO₂ and promoted interfacial reactions, thereby facilitating the wetting and spreading of AgCuInTi on the composites' surface, ultimately reducing the contact angle from 71° to 33°. The typical microstructure of the joint was characterized as: α-Al₂O₃/Ti₂O + Cu₃Ti₃O/Ag(s,s) + Cu(s,s)/Cu₃Ti₃O + Ti₅Si_x (x = 2, 3, and 4) + Na₂Ti₉O₁₉/SiO₂f/SiO₂, achieving a gradient transition in the coefficient of thermal expansion across the brazed seam. Therefore, a maximum shear strength of 52 MPa was achieved when the joint was brazed at 820 °C for 10 min with a corrosion time of 10 min, which is 2.5 times higher than that of the joint without surface modification.

Graphical abstract

3. Results and discussion

4. Conclusions

CRediT authorship contribution statement

Declaration of competing interest

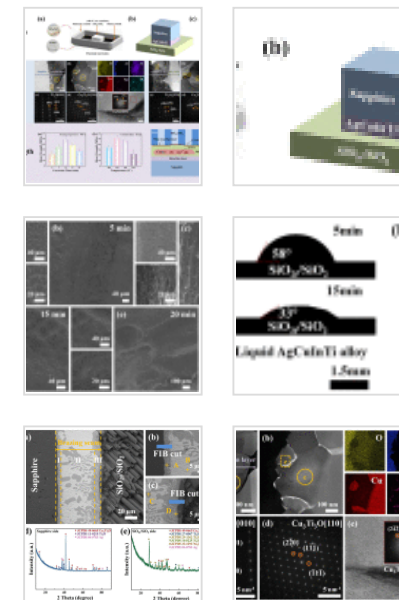
Acknowledgements

Appendix. Supplementary materials

References

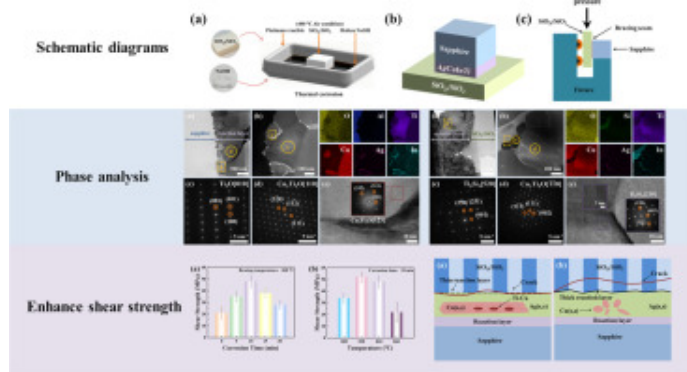
[Show full outline](#) ▼

Figures (13)



[Show 7 more figures](#) ▼

Tables (2)



Download: [Download high-res image \(267KB\)](#)

Download: [Download full-size image](#)

Keywords

SiO₂f/SiO₂ composites; Sapphire; Brazed-joints; Wettability; Thermal corrosion; Microstructures

[Previous article in this issue](#)

[Next article in this issue](#)

1. Introduction

SiO₂f/SiO₂ composites exhibit high microwave transmittance, coupled with superior properties such as high melting point, excellent ablation resistance, and low thermal conductivity, making them ideal candidates for next-generation radome materials [1], [2], [3]. Nevertheless, as radar technology advances, there is an increased demand for high transmittance across visible and infrared bands in addition to microwaves [4]. The conventional SiO₂f/SiO₂ composites can no longer meet these multi-band transmittance requirements for radome materials [5,6]. Sapphire, namely single-crystalline aluminum oxide (α -Al₂O₃), exhibits exceptionally high transmittance across visible, near-infrared,

[Table 1. EDS analysis of spots marked in ...](#)

[Table 2. EDS analysis of spots marked in ...](#)

Extras (1)

[Document](#)

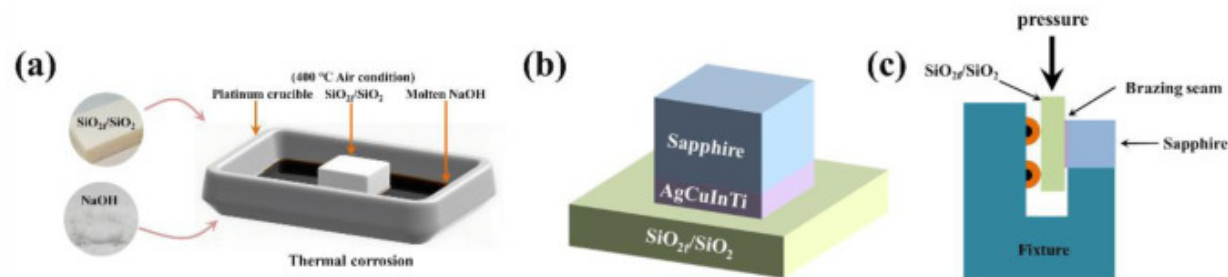
and mid-infrared spectra, demonstrating significant complementarity with SiO_{2f}/SiO₂ composites [7,8]. This characteristic makes the effective joining of sapphire and SiO_{2f}/SiO₂ crucial for manufacturing novel multi-spectral transmissive radar radomes. Various techniques exist for joining these heterogeneous materials, including adhesive bonding, riveting, and brazing [[9], [10], [11]]. However, riveting induces localized stress concentration while substantially increasing structural mass, and adhesive bonding suffers from poor high-temperature resistance coupled with low joint strength [12,13]. In contrast, brazing offers superior joint strength and excellent high-temperature resistance, and eliminates mechanical stress concentration, making it a promising approach for bonding heterogeneous materials [[14], [15], [16]].

Two challenges must be overcome to achieve reliable joining between SiO_{2f}/SiO₂ composites and sapphire: (1) wettability issues with SiO_{2f}/SiO₂ composites: the unique surface morphology and extremely low surface energy of these composites hinder the wetting and spreading of brazing filler on their surfaces [17,18]. (2) Thermal expansion mismatch: the coefficient of thermal expansion (CTE) of SiO_{2f}/SiO₂ composites differs by an order of magnitude from that of sapphire (CTE_{sapphire}: $5.3 \times 10^{-6} \text{ K}^{-1}$; CTE_{SiO_{2f}/SiO₂}: $0.6 \times 10^{-6} \text{ K}^{-1}$), leading to the generation of excessive residual stresses during cooling [19]. Previous studies have proposed various solutions to these challenges [20]. Under constant brazing temperature conditions, the residual stresses formed post-brazing increase with the magnitude of thermal expansion coefficient mismatch at the interface [21,22]. Therefore, Wang et al. [23] incorporated negative thermal expansion particles into the joint to reduce the CTE of the brazing seam, thereby mitigating residual stresses. On the other hand, Ba et al. [24] employed selective corrosion to remove poorly wettable phases from the composite, exposing higher-surface-energy phases to enhance filler metal wettability. Nevertheless, emerging challenges arise with next-generation SiO_{2f}/SiO₂ composites where both fiber reinforcement and matrix maintain amorphous structures with homogeneous chemical characteristics, coupled with inherently poor wettability to active brazing filler. Current surface modification strategies fall short in addressing this challenge.

In this study, surface modification of SiO_{2f}/SiO₂ composites was achieved through NaOH thermal corrosion. This dual-function surface engineering strategy can simultaneously alter the unique surface morphology of SiO_{2f}/SiO₂ and generate reactive chemical species at the interface, thereby synergistically enhancing wettability. Then, sapphire and the SiO_{2f}/SiO₂ were joined using AgCuInTi filler material, owing to its high reactivity, low brazing temperature, superior plasticity, and toughness. The effects of process parameters on the mechanical properties and microstructural evolution of brazed joints were systematically investigated. By regulating interfacial reactions, a graded transition in the CTE of the brazed seam was achieved, effectively mitigating residual stresses. Additionally, the fracture mechanisms of joints under various conditions were analyzed.

2. Experimental section

Fig. 1(a) illustrates the schematic of NaOH thermal corrosion. Initially, the SiO_{2f}/SiO₂ composites were sectioned into 10 mm × 10 mm × 5 mm. Subsequently, 0.2 g of NaOH pellets and SiO_{2f}/SiO₂ were placed into a platinum crucible, which was then transferred to a muffle furnace. The temperature was heated to 400 °C at a rate of 10 °C/min, followed by holding times of 5, 10, 15, and 20 min, respectively. After thermal corrosion, SiO_{2f}/SiO₂ underwent alternating ultrasonic cleaning using deionized water and ethanol.



[Download: Download high-res image \(168KB\)](#)

[Download: Download full-size image](#)

Fig. 1. Schematic diagrams of (a) NaOH thermal corrosion, (b) brazing joints assembly, and (c) shear strength test.

Sapphire substrates were cut into 5 mm × 5 mm × 5 mm cubes and cleaned with ethanol. The brazing filler employed in this paper was a 100 μm-thick AgCuInTi foil with solidus and liquidus temperatures of 610 and 710 °C, respectively. Following the sequence shown in Fig. 1(b), the sapphire, brazing foil, and SiO_{2f}/SiO₂ were assembled and placed into a vacuum furnace for brazing. After achieving a vacuum of less than 5×10^{-3} Pa, the temperature was increased to 800–860 °C at 10 °C/min, held for 10 min, and then decreased to room temperature at 5 °C/min.

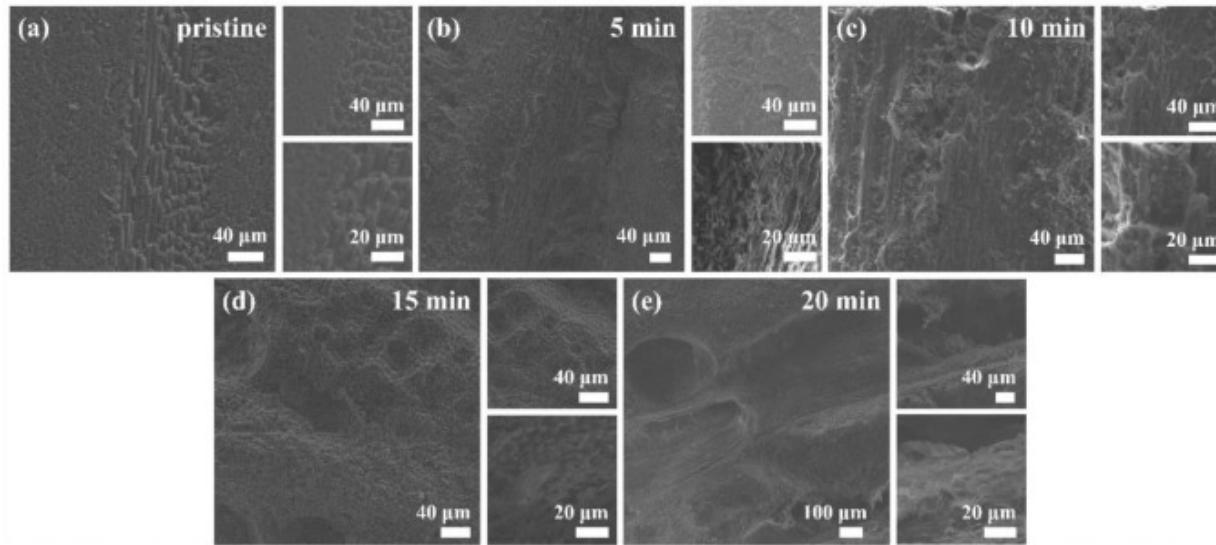
The microstructural morphology of the specimens was characterized using a scanning electron microscope (SEM, Zeiss Merlin Compact), coupled with energy dispersive spectroscopy (EDS) for elemental composition analysis. In this paper, within all SEM images of the joints, the left side corresponds to sapphire, while the right side represents the SiO_{2f}/SiO₂. The brazed joint was etched using a HNO₃ solution. The etching process was continued until the reaction layer was exposed. Subsequently, micro-zone X-ray diffraction (XRD, Empyrean) analysis was conducted on both reaction layers. The focused ion beam samples were prepared using an electron-ion dual-beam system (Helios 5 UX) and subsequently characterized by transmission electron microscope (TEM, Talos F200X) for structural observation and analysis. The shear properties of the brazed joints were evaluated via a universal testing machine (Instron 5982), with five specimens tested per group to ensure statistical reliability.

3. Results and discussion

3.1. Surface activation of SiO_{2f}/SiO₂ composites by thermal corrosion

Fig. 2 illustrates the comparative surface of SiO_{2f}/SiO₂ composites before and after thermal corrosion. The pristine SiO_{2f}/SiO₂ composites present a structure of smooth silica fibers embedded within a continuous silica matrix, as shown in Fig. 2(a). As demonstrated in Fig. 2(b–e), after thermal corrosion for 5–20 min, the surface of SiO_{2f}/SiO₂ composites changes from the initial flat state to uneven. Both the quartz matrix and fibers on the surface undergo partial etching; the corrosion behavior exhibits no

significant selectivity. This phenomenon arises from the identical chemical composition of fibers and matrix, enabling molten NaOH to concurrently react with both components. With the increase in corrosion time, a progressive consumption of SiO_{2f}/SiO₂ composites is observed, and voids emerge when the holding time reaches 20 min (Fig. 2(e)).

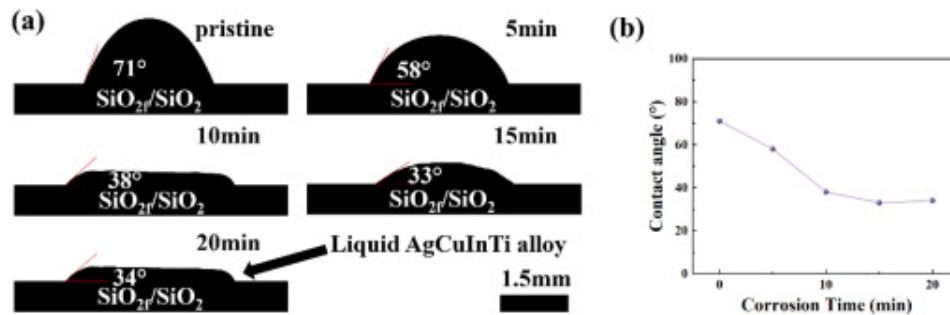


[Download: Download high-res image \(525KB\)](#)

[Download: Download full-size image](#)

Fig. 2. The surface morphology of SiO_{2f}/SiO₂ composites after thermal corrosion: (a) 0 min, (b) 5 min, (c) 10 min, (d) 15 min, and (e) 20 min.

The wettability of AgCuInTi filler on SiO_{2f}/SiO₂ composites was investigated at 820 °C for 10 min, as shown in Fig. 3. Due to the synergistic effects of its unique surface architecture and the intrinsic chemical inertness of silica, the pristine SiO_{2f}/SiO₂ composites exhibit a contact angle of 71°. The contact angle of the brazing filler on the etched SiO_{2f}/SiO₂ composites surface exhibits a progressive reduction from 71° to a minimum of 33°, corresponding to a 54 % decrease.



[Download: Download high-res image \(152KB\)](#)

[Download: Download full-size image](#)

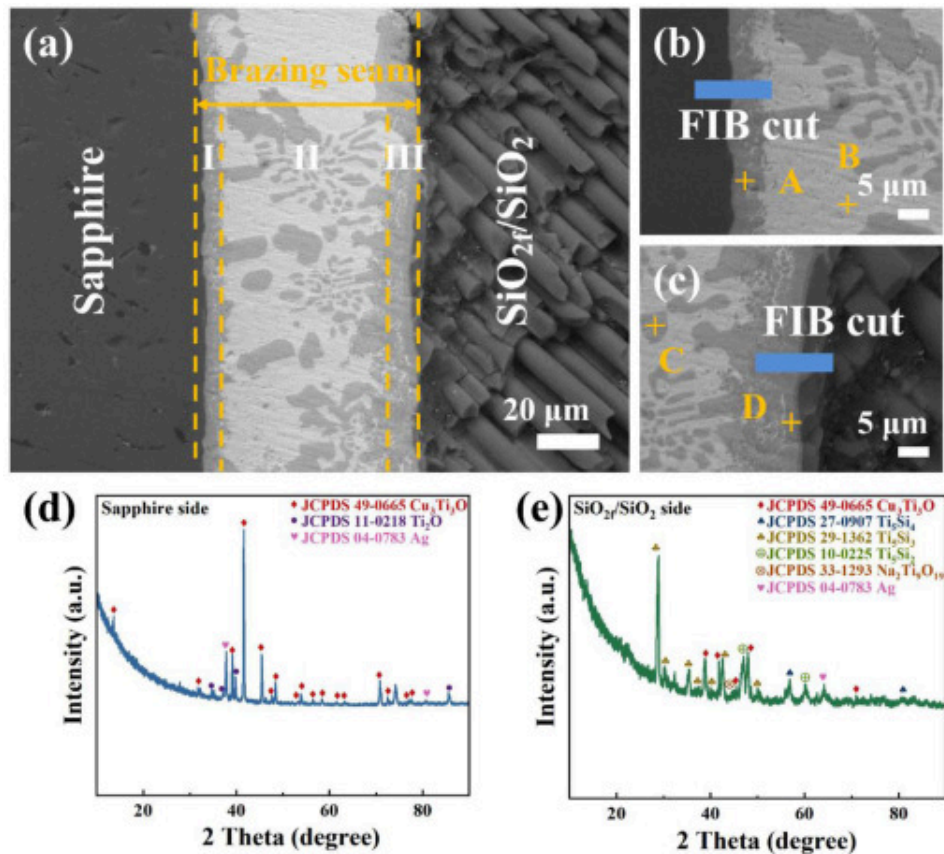
Fig. 3. Wetting experiment of AgCuInTi filler on the SiO_{2f}/SiO₂ surface: (a) wetting images of AgCuInTi filler; (b) the effect of corrosion time on the contact angle.

This significant enhancement can be attributed to two synergistic mechanisms. First is the alteration in surface morphology, as shown in Fig. S3 in the Supplementary Materials. Although the original composite appears smooth, gaps actually persist between fibers and between fibers and the matrix. These cliff-like discontinuities impede the formation of continuous reaction layers. Additionally, due to the high surface tension, molten brazing filler cannot spread into these gaps, making it impossible to form a pinning effect. In contrast, after thermal treatment, a continuous Na₂SiO₃ layer forms on the surface, filling these gaps and enabling the brazing filler to diffuse smoothly.

Second is the increased surface energy. Previous studies show that the SiO_{2f}/SiO₂-AgCuTi-based filler interface is a typical reactive wetting interface [25], where the spreading rate is controlled by the growth rate of the interfacial reaction layer [26,27]. The inherently low surface energy of amorphous SiO₂ retards interfacial reactions between the matrix and brazing filler [28]. The Na₂SiO₃ layer, however, exhibits much higher surface energy compared to amorphous SiO₂, which accelerates the interface reaction rate. This allows more active elements in the brazing filler to be consumed and promotes their faster diffusion to the interface, driven by chemical potential. Consequently, this results in a lower equilibrium wetting angle.

3.2. Typical microstructure of the sapphire-SiO_{2f}/SiO₂ joint

Fig. 4 presents the typical microstructure of the sapphire-SiO_{2f}/SiO₂ joint. The brazing parameters were 820 °C/10 min, with pre-brazing thermal corrosion of the SiO_{2f}/SiO₂ at 400 °C/10 min. The joint interface shows no obvious defects and comprises three distinct regions: Region I (reaction layer on the sapphire side), Region II (Ag-based and Cu-based solid solutions), and Region III (reaction layer on the SiO_{2f}/SiO₂ side).



Download: [Download high-res image \(694KB\)](#)

Download: [Download full-size image](#)

Fig. 4. Microstructure characterization of the sapphire-SiO_{2f}/SiO₂ joint: (a–c) SEM images of the joint; (d) XRD pattern on the sapphire side; (e) XRD pattern on the SiO_{2f}/SiO₂ side.

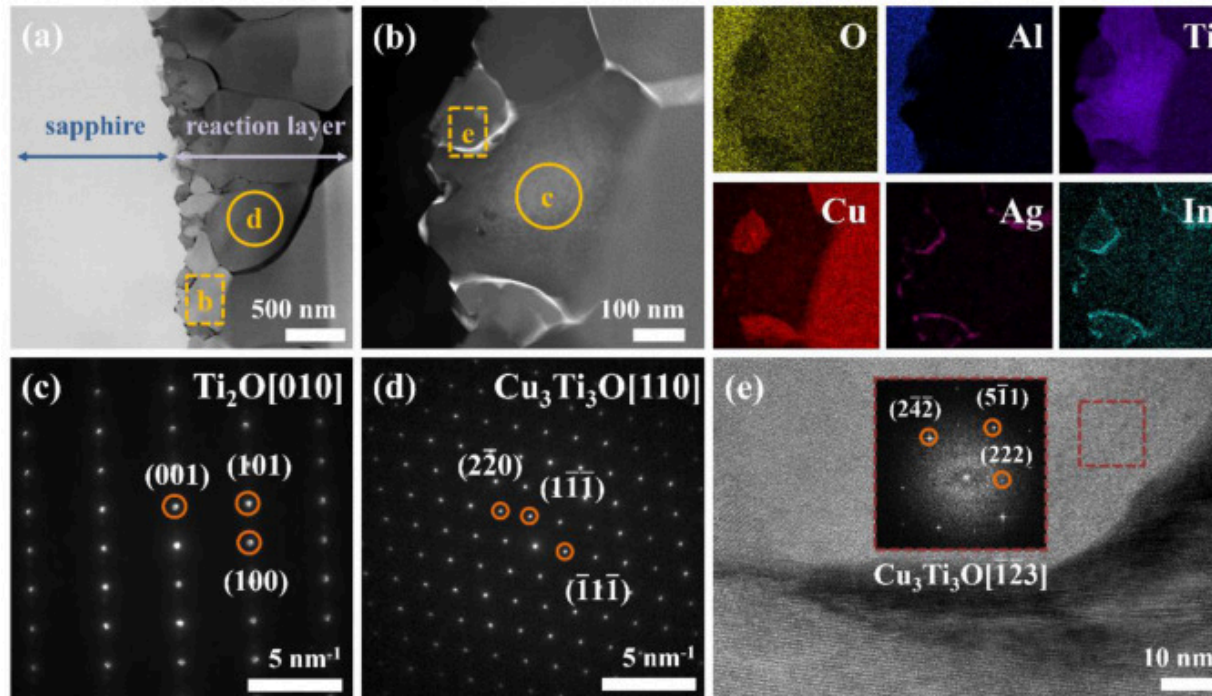
Combined with the XRD pattern in Fig. 4(d) and EDS analysis in Table 1, the reaction products in Region I can be identified as Ti-O and Ti-Cu-O compounds, indicating that Ti plays a critical role in interfacial reactions on the sapphire side. Dissolution of the O element from sapphire and subsequent reaction with Ti are key steps for reactive wetting. Region II exhibits a darker Cu-based solid solution and a lighter Ag-based solid solution, with no Cu-Ti intermetallic found. In Region III, the XRD pattern (Fig. 4(e)) reveals complex phase compositions, including multiple Ti-Si compounds, Cu-Ti-O phases, and possible Na-Ti-O phases.

Table 1. EDS analysis of spots marked in Fig. 4 (at.%).

Spots	O	Al	Si	Ti	Cu	Ag	In	Possible phase
A	12.2	1.9	–	35.3	48.5	1.1	1.0	Ti ₂ O + Ti ₃ Cu ₃ O
B	10.6	–	–	–	18.2	62.6	8.6	Ag(s,s)
C	5.6	–	–	1.4	88.4	3.1	1.5	Cu(s,s)
D	34.3	–	18.6	43.5	2.5	0.8	0.3	Ti ₅ Si ₃ + Ti ₃ Cu ₃ O

To further analyze reaction products and investigate the mechanism of thermal corrosion on SiO₂/SiO₂ wettability, TEM characterization was performed on Regions I and III, as shown in Fig. 5, Fig. 6, Fig. 7. Fig. 5(a) shows that the interfacial reaction layer near the sapphire side consists of fine grains (< 100 nm), with grain size increasing toward the brazing filler side. Selective area electron diffraction (SAED) patterns (Fig. 5(c and d)) confirm that phase d in Fig. 5(a) corresponds to Cu₃Ti₃O, while phase c in Fig. 5(b) is identified as Ti₂O. Elemental mapping and fast Fourier transform (FFT) analysis (Fig. 5(e)) further verify that the fine grains adjacent to the sapphire are Cu₃Ti₃O. Therefore, it can be concluded that during the brazing process, the Ti element within the filler preferentially reacts with O from sapphire, forming Ti₂O nanocrystals. As the reaction proceeds, Ti₃Cu₃O compounds nucleate and grow upon the Ti₂O layer, ultimately

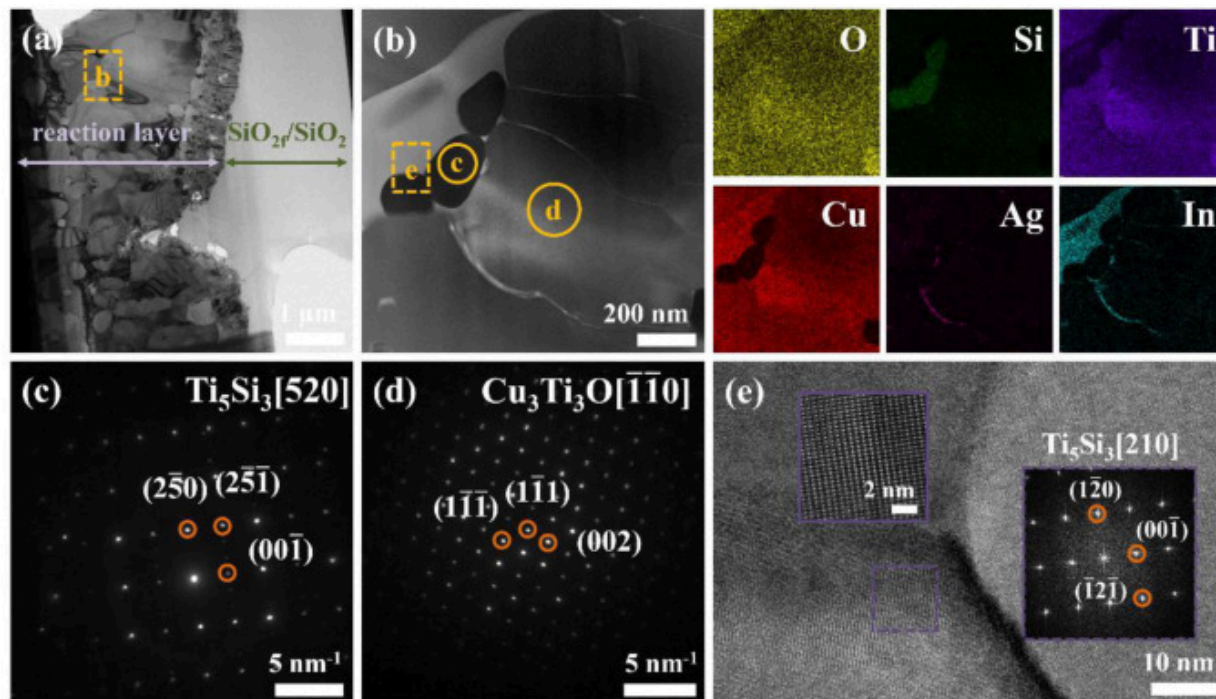
displacing the Ti₂O phase while simultaneously promoting the wetting and spreading behavior of the molten filler at the interface. The corresponding reaction equations are provided below:



[Download: Download high-res image \(956KB\)](#)

[Download: Download full-size image](#)

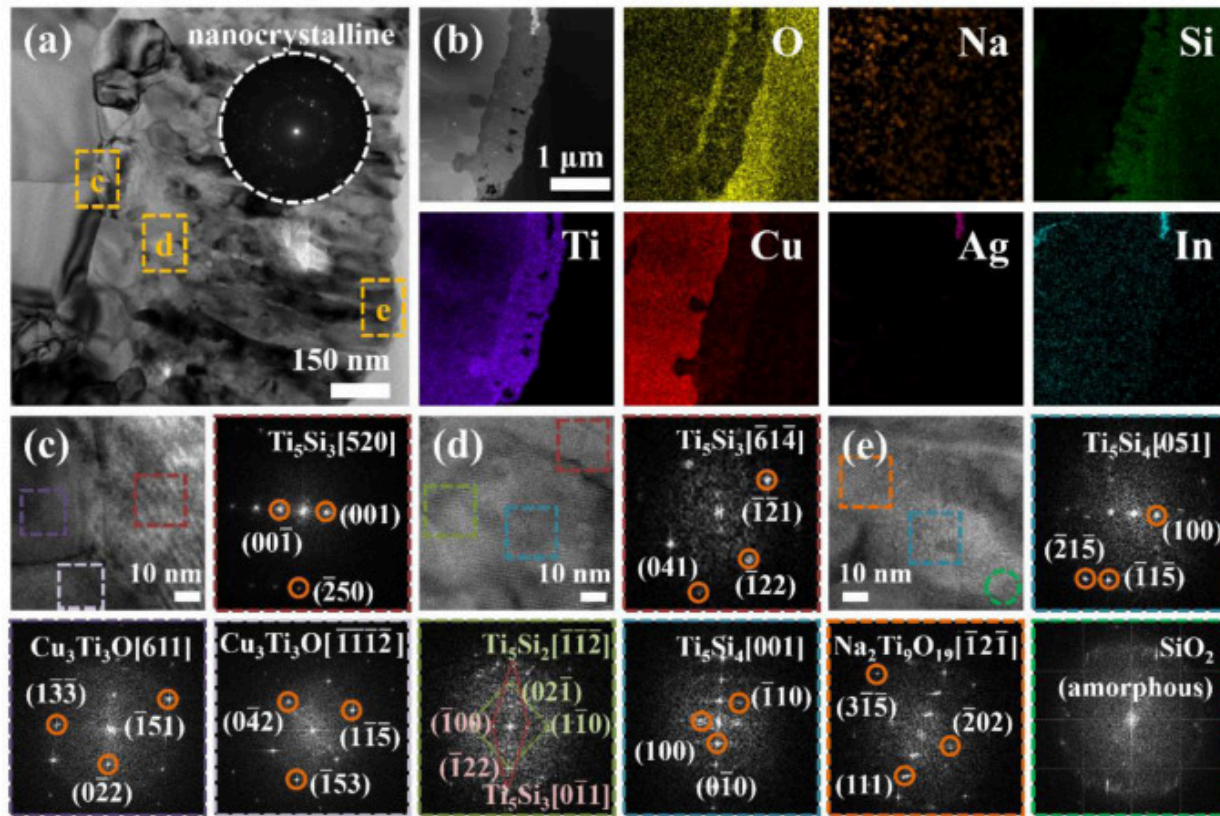
Fig. 5. TEM analysis of the reaction layer on the sapphire side: (a) bright field (BF) image of the interface; (b) elemental mapping; (c) SAED of phase c in (b); (d) SAED of phase d in (a); (e) high resolution TEM (HRTEM) and FFT of region e in (b).



[Download: Download high-res image \(949KB\)](#)

[Download: Download full-size image](#)

Fig. 6. TEM analysis of the reaction layer on the SiO_{2f}/SiO₂ side: (a) BF image of the interface; (b) elemental mapping; (c) SAED of phase c in (b); (d) SAED of phase d in (b); (e) HRTEM and FFT of region e in (b).



Download: [Download high-res image \(1MB\)](#)

Download: [Download full-size image](#)

Fig. 7. TEM analysis of the interface of the reaction layer on the SiO₂f/SiO₂ side: (a) BF image of the nanocrystalline; (b) elemental mapping; (c) HRTEM and FFT of region c in (a); (d) HRTEM and FFT of region d in (a); (e) HRTEM and FFT of region e in (a).

The overall morphology of the reaction layer on the SiO₂f/SiO₂ side (Fig. 6(a)) comprises a thin nanograined layer near the composites and a thicker inner layer. Analysis of the inner layer (Fig. 6(c–e)) reveals that the three smaller grains in Fig. 6(b) are Ti₅Si₃, while the larger grain is Cu₃Ti₃O. Notably, the formation of Ti₅Si₃ at distances > 2.5 μm from the SiO₂f/SiO₂ substrates is unprecedented in prior studies [[29], [30], [31], [32]], due to limited Si diffusion in amorphous SiO₂ [33]. The presence of Ti₅Si₃ in Fig. 6(b) is ascribed

to in situ Na₂SiO₃ generated on the SiO_{2f}/SiO₂ surface during pre-treatment. This indirectly demonstrates that the alteration of chemical properties on SiO_{2f}/SiO₂ surfaces induced by thermal corrosion treatment contributes to the improved wettability.

Fig. 7(a) shows the BF image and diffraction patterns of the nanocrystalline reaction layer near the SiO_{2f}/SiO₂ composites side, exhibiting typical polycrystalline characteristics. Elemental mapping in Fig. 7(b) shows that the nanocrystalline region is rich in Ti and Si, with some Na. Dividing this area into regions c (near brazing seam), d (nanograins), and e (near SiO_{2f}/SiO₂) for further analysis, Fig. 7(c) shows two Cu₃Ti₃O grains in region c, with Ti₅Si₃ at their interfaces. FFT results in Fig. 7(d) indicate the nanocrystalline region comprises multiple reaction products, such as Ti₅Si₄, Ti₅Si₃, and Ti₅Si₂. The generation of various Ti-Si phases suggests improved composite wettability as Ti-Si compounds consume less Ti compared to Ti₂O or Cu₃Ti₃O, maintaining brazing filler activity. As shown in Fig. 7(e) and S4, a novel Na₂Ti₉O₁₉ phase is observed. This reaction product consumes a substantial amount of Ti, creating a locally Si-rich microzone around it. Consequently, the Ti₅Si₄ phase can be observed adjacent to the Na₂Ti₉O₁₉ phase. Combined with the previously mentioned Ti₅Si₃ generation far from the substrate, this suggests a reaction sequence: during brazing, Na₂SiO₃ first reacts with the filler metal to form Ti₂O, Na₂O, and Si. Subsequently, the generated Si further reacts with the filler metal to produce Ti₅Si_x. As the brazing filler further wets and spreads, Na₂O in the alloy reacts with sufficient SiO₂ to generate Na₂Ti₉O₁₉ and Ti-Si phases. The reactions are as follows:



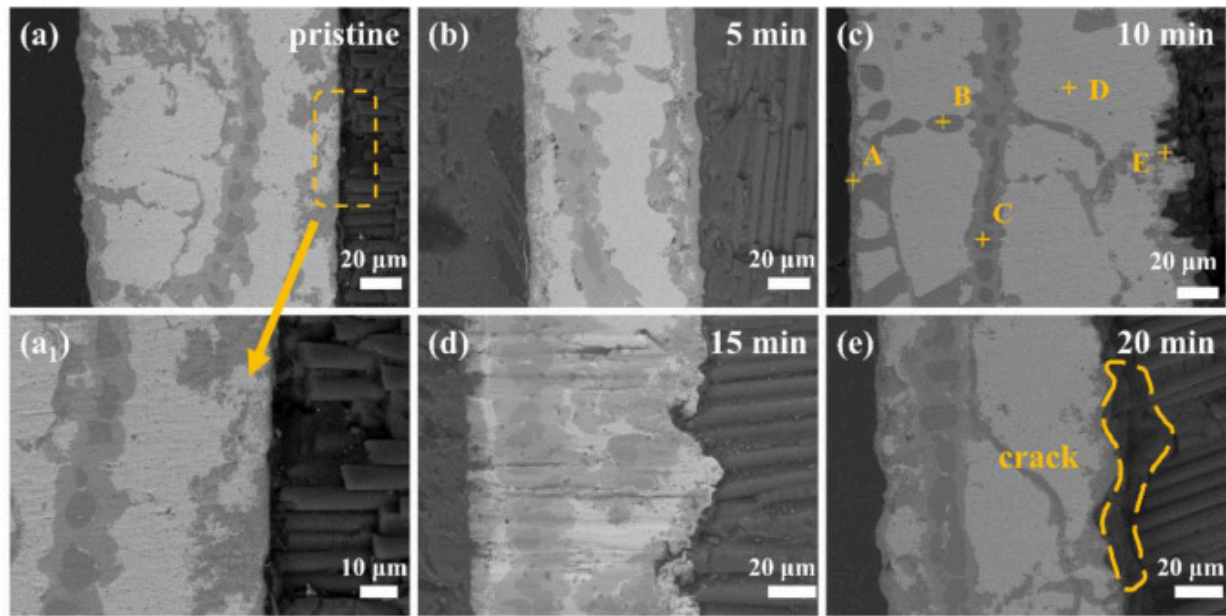
Based on the above analysis, the interfacial microstructure of sapphire-SiO_{2f}/SiO₂ joint is $\alpha\text{-Al}_2\text{O}_3/\text{Ti}_2\text{O} + \text{Cu}_3\text{Ti}_3\text{O}/\text{Ag}(\text{s},\text{s}) + \text{Cu}(\text{s},\text{s})/\text{Cu}_3\text{Ti}_3\text{O} + \text{Ti}_5\text{Si}_x$ ($x = 2, 3, \text{ and } 4$) +

Na₂Ti₉O₁₉/SiO_{2f}/SiO₂. Notably, this structure mitigates the CTE mismatch between the substrate and brazing filler. The CTEs are as follow: AgCuInTi ($21 \times 10^{-6} \text{ K}^{-1}$), Cu₃Ti₃O ($12.8 \times 10^{-6} \text{ K}^{-1}$) [34], Ti₂O ($8.6 \times 10^{-6} \text{ K}^{-1}$), α -Al₂O₃ ($5.3 \times 10^{-6} \text{ K}^{-1}$), Ti₅Si₃ ($2.8 \times 10^{-6} \text{ K}^{-1}$) [35], and SiO_{2f}/SiO₂ ($0.6 \times 10^{-6} \text{ K}^{-1}$). The residual stress was measured using XRD, as shown in Fig. S5. It can be observed that the residual stress of the directly brazed joint is as high as -339.30 MPa, while the residual stress of the joint brazed after 10 min of thermal corrosion is only -171.30 MPa, representing a decrease of 49.5 %. The finite element simulation results presented in Fig. S6 also exhibit the same trend. The gradual CTE transition between phases in the weld can alleviate residual stress, thereby improving joint stability.

3.3. Effect of parameter on the interfacial microstructure of sapphire-SiO_{2f}/SiO₂ joint

The SiO_{2f}/SiO₂ composites under different corrosion time and sapphire were brazed at 840 °C/10 min, and the interfacial microstructure of joints were shown in Fig. 8. A thin pale gray reaction layer is consistently observed at the sapphire/AgCuInTi interface across all parameter sets, indicating that the brazing filler achieved sufficient spreading and wetting on the sapphire surface, thereby establishing a sound metallurgical bond. As shown in Fig. 8(a), the interfacial reaction layer on the SiO_{2f}/SiO₂ side exhibits discontinuous morphology with heterogeneous distribution, resulting in gaps at the interface (Fig. 8(a₁)). This phenomenon originates from the loose textures of the composites, which restrict the continuous reaction between the brazing filler and the substrate within the limited time, consistent with the previous wetting experiment. As illustrated in Fig. 8(b–d), with the increase of corrosion time, it can be seen that the brazing filler gradually infiltrates into the SiO_{2f}/SiO₂ composites, forming a continuous reaction layer that facilitates metallurgical bonding between the SiO_{2f}/SiO₂ and brazing seam. However, when the corrosion duration extends to 20 min, continuous cracks develop at the interface due to excessive brazing filler infiltration into the substrate, coupled with the mismatch of thermal coefficient of expansion between the AgCuInTi and SiO_{2f}/SiO₂ composites. This mismatch induces significant residual stresses at the

SiO₂_f/SiO₂/AgCuInTi interface during cooling, which can be mitigated by subsequent adjustment of brazing parameters to regulate stress distribution. In addition, island-like darker zones observed within the Cu(s,s) in Fig. 8(c) are identified as Cu₄Ti₃ through EDS analysis (Table 2). The In element demonstrates uniform distribution across the brazing seam, mostly dissolved in the Ag-based solid solution [22,36].



Download: [Download high-res image \(661KB\)](#)
Download: [Download full-size image](#)

Fig. 8. The microstructure of sapphire-SiO₂_f/SiO₂ joints: (a and a₁) pristine; (b) 5 min; (c) 10 min; (d) 15 min; (e) 20 min.

Table 2. EDS analysis of spots marked in Fig. 8(c) (at.%).

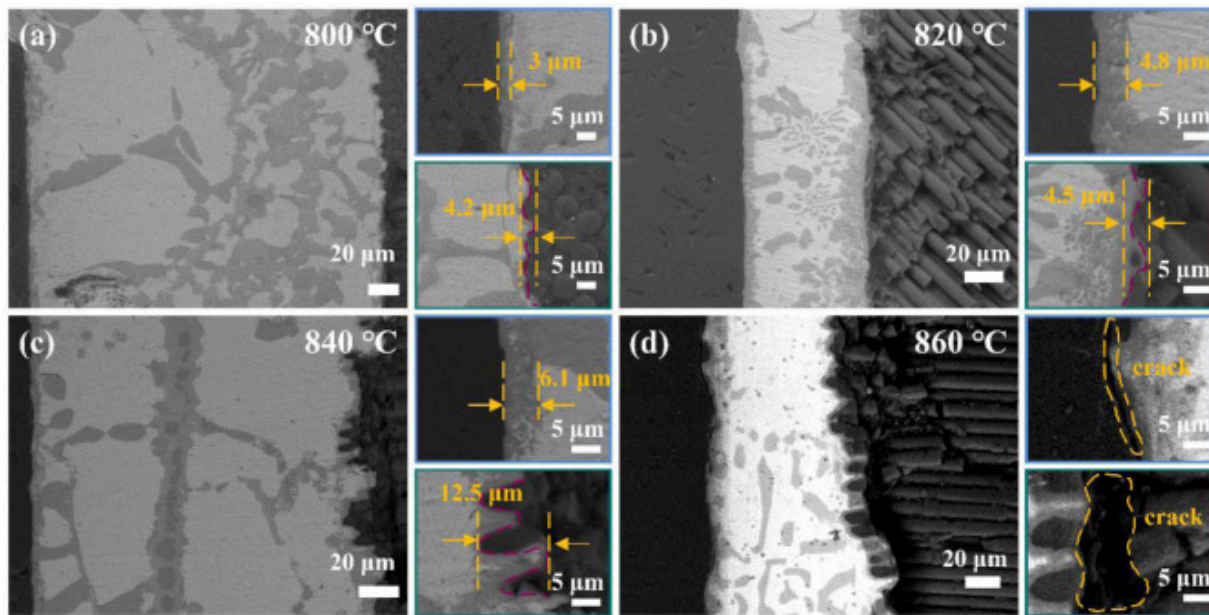
Spots	O	Al	Si	Ti	Cu	Ag	In	Possible phase
A	25.2	1.7	0.6	31.5	28.9	2.3	3.9	TiO + Ti ₃ Cu ₃ O
B	2.3	–	–	0.7	92.7	2.3	2.0	Cu(s,s)

Spots	O	Al	Si	Ti	Cu	Ag	In	Possible phase
C	8.8	–	–	40.6	49.5	1.1	–	Cu ₄ Ti ₃
D	12.1	–	–	–	6.6	71.1	10.2	Ag(s,s)
E	32.9	–	8.3	32.8	24.9	0.8	0.3	Ti ₅ Si ₃ + Ti ₃ Cu ₃ O

To determine the optimal corrosion time for SiO₂f/SiO₂ composites, shear strength tests were conducted on the sapphire-SiO₂f/SiO₂ joints. As shown in Fig. S7, after considering the trade-off between surface activation of the SiO₂f/SiO₂ and potential damage under different corrosion durations, a corrosion time of 10 min was selected as the optimal parameter, at which the joint achieved the highest shear strength. In subsequent experiments, all SiO₂f/SiO₂ composites were subjected to a 10-minute NaOH thermal treatment prior to brazing.

Subsequently, the effect of varying brazing temperatures (800–860 °C) on the microstructural morphology of sapphire-SiO₂f/SiO₂ joints was systematically investigated, as comprehensively detailed in Fig. 9. The reaction layer thickness at the sapphire/AgCuInTi interface was initially measured. The results demonstrate that the thickness exhibits a progressive increase with rising temperature, increasing from 3 μm at 800 °C to 6.1 μm at 840 °C. Consistently, observation of the SiO₂f/SiO₂ composites/AgCuInTi interface reveals a progressive widening of the infiltration zone formed by the brazing filler penetrating into the substrate as the temperature increases. This phenomenon primarily stems from the elevated brazing temperature, enhancing the diffusion of Ti into the substrate, which subsequently intensifies interfacial reactions. However, at the brazing temperature of 860 °C, the interfaces between the filler metal and both substrates exhibit significant cracks, attributed to the excessive brazing temperature exacerbating residual stresses. Furthermore, at lower brazing temperatures, as illustrated in Fig. 9(a and b), no dark island-shaped brittle phases formed at 840 °C are observed. This demonstrates that modulating the brazing temperature can effectively

inhibit Ti-Cu interfacial reactions, thereby reducing the formation of intermetallic compounds.



Download: [Download high-res image \(792KB\)](#)

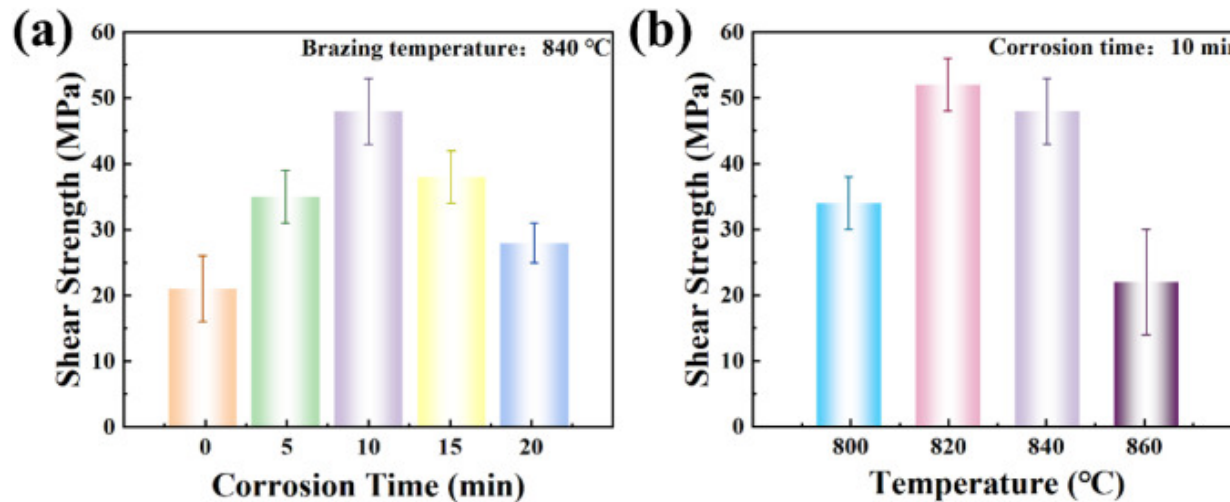
Download: [Download full-size image](#)

Fig. 9. The microstructure of sapphire-SiO_{2f}/SiO₂ joints at different temperatures: (a) 800 °C; (b) 820 °C; (c) 840 °C; (d) 860 °C.

3.4. Relationships between mechanical performance and microstructure of the sapphire-SiO_{2f}/SiO₂ brazed joints

Fig. 10 illustrates the shear strength of brazed joints under varying process parameters. As shown in Fig. 10(a), surface activation of SiO_{2f}/SiO₂ prior to brazing substantially enhances the joint strength, with shear strength progressively increasing with corrosion time until reaching a peak of 48 MPa at 10 min, followed by a gradual decline. Based on Fig. 4, during direct brazing of sapphire to SiO_{2f}/SiO₂, the filler metal predominantly establishes a continuous reaction layer along the sapphire interface, whereas the

SiO₂f/SiO₂ demonstrates fragmented reaction layer morphology with localized unbonded regions. Surface activation treatment enhances the degree of interface reaction, strengthens the metallurgical bonding between the SiO₂f/SiO₂ composites and brazing seam, and thereby improves the mechanical properties of the joints. However, prolonged corrosion treatment (20 min) induces void formation within the composite substrate (Fig. 2(e)), compromising its interfacial strength and ultimately degrading the mechanical properties of the joints.



Download: [Download high-res image \(349KB\)](#)

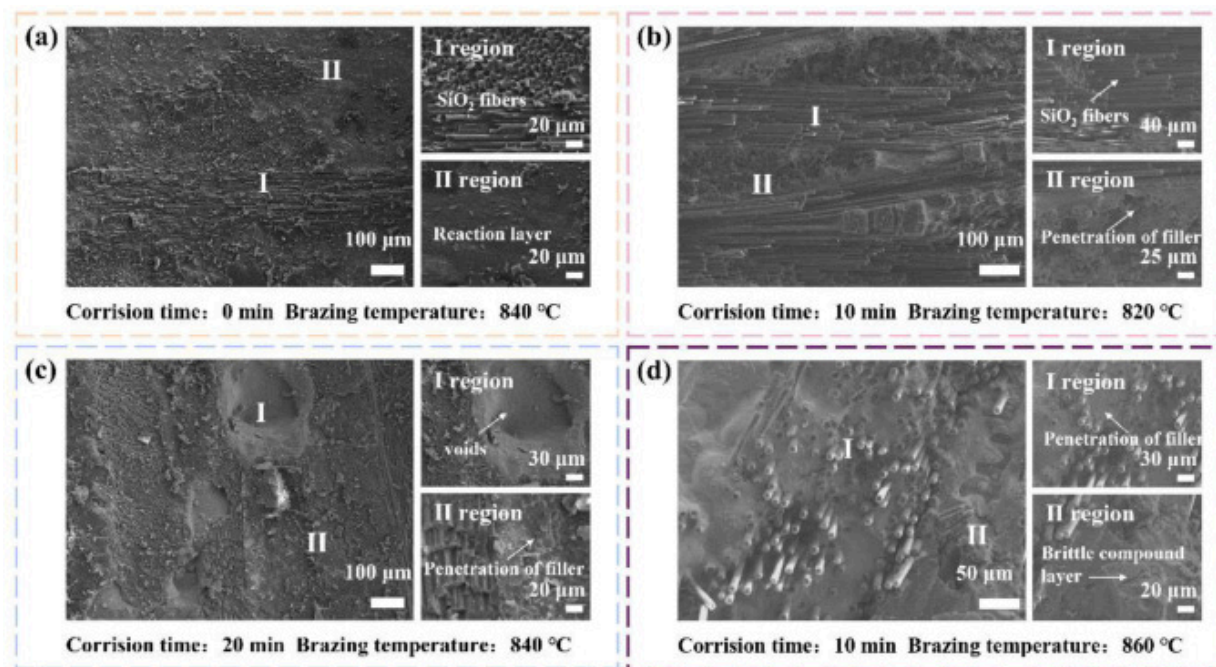
Download: [Download full-size image](#)

Fig. 10. Effect of process parameters on the shear strength of the brazed joints: (a) corrosion time; (b) brazing temperature.

Fig. 10(b) shows the shear strength of the joint at different brazing temperatures, with a trend similar to the corrosion time. As the brazing temperature increases, the shear strength exhibits an initial ascending trend, peaking at 52 MPa at 820 °C, followed by a subsequent decline. As evidenced by Fig. 9(d), excessive brazing temperature leads to the formation of Cu₄Ti₃ brittle intermetallic within the brazing seam and induces interfacial

crack formation at the brazed seam/substrate interface, thereby deteriorating the mechanical strength of the joints.

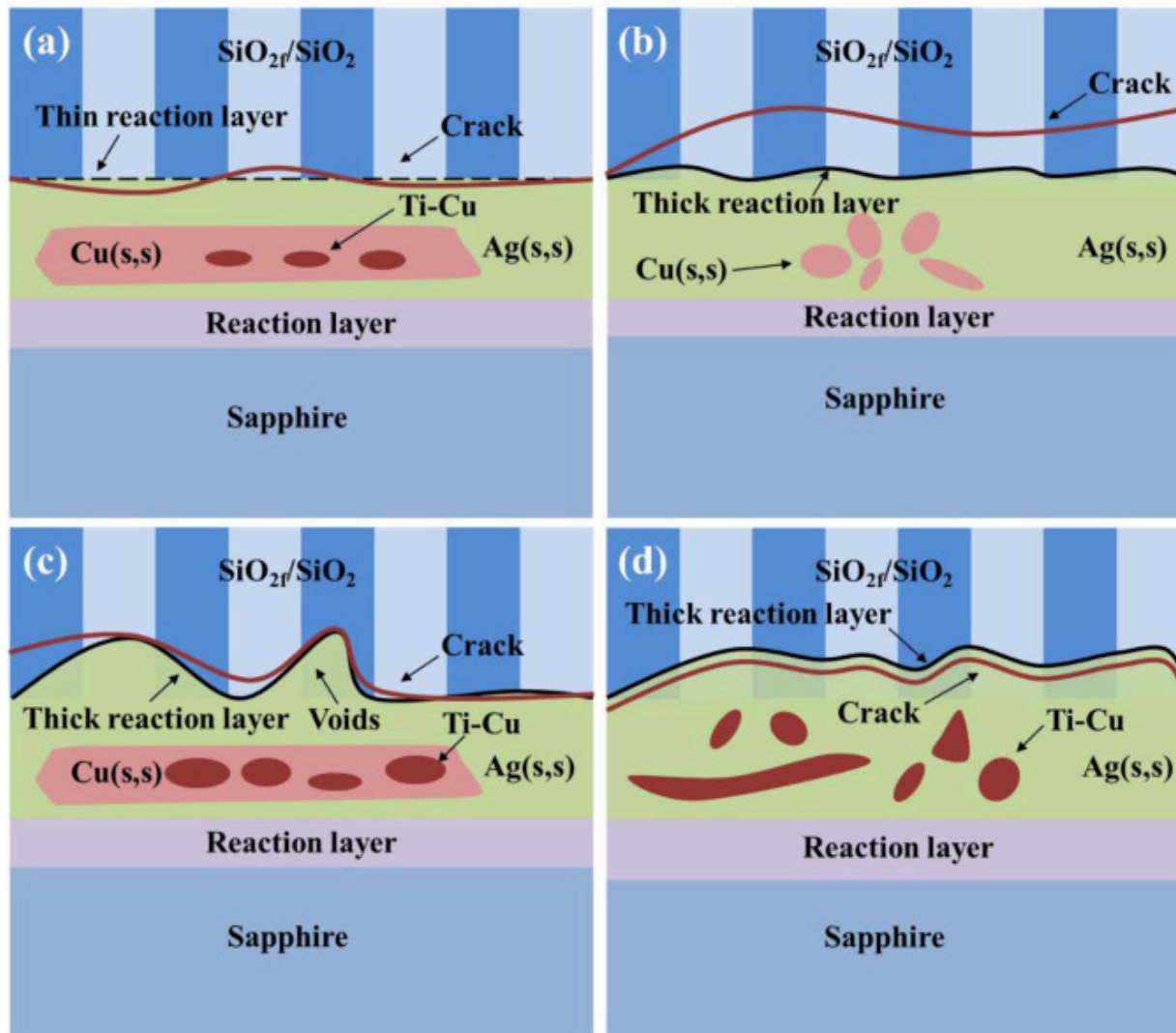
Fig. 11 presents four typical fracture morphologies of sapphire-SiO_{2f}/SiO₂ joints on the sapphire side, respectively, while Fig. 12 illustrates the corresponding fracture mechanism for each morphology. The EDS results and composition analysis of each fracture surface are presented in Figs. S9–S12 and Tables S2–S5. As shown in Fig. 11(a), when directly brazing sapphire and SiO_{2f}/SiO₂, the joint fracture primarily presents a small amount of silica matrix (I) and the reaction layer on the sapphire side (II), indicating that the fracture occurs at the SiO_{2f}/SiO₂/ brazing seam interface. As shown in Fig. 12(a), this fracture behavior originates from the restricted bonding between the brazing seam and SiO_{2f}/SiO₂, with cracks propagating predominantly through the unbonded area.



[Download: Download high-res image \(916KB\)](#)

[Download: Download full-size image](#)

Fig. 11. The fracture morphology of the sapphire-SiO_{2f}/SiO₂ joints on the sapphire side: (a) directly joining at 840 °C; (b) joining at 820 °C after 10 min of corrosion; (c) joining at 840 °C after 20 min of corrosion; (d) joining at 860 °C after 10 min of corrosion.



Download: [Download high-res image \(804KB\)](#)

Download: [Download full-size image](#)

Fig. 12. The fracture mechanism of (a) Fig. 11(a); (b) Fig. 11(b); (c) Fig. 11(c); (d) Fig. 11(d).

After surface activation, the fracture morphology exhibits extensive fiber pull-out traces, indicating that the fibers participate in the fracture process and play a role in inhibiting crack propagation, also demonstrating the strong bonding between the brazing seam and SiO_{2f}/SiO₂, as shown in Figs. 11(b) and S13. Furthermore, multi-interfacial SiO₂ fiber fracture can be observed, demonstrating that fracture does not occur along a single interface [37,38]. This multi-interface fracture behavior directly contributes to the enhanced joint strength. The corresponding fracture mechanism is schematically illustrated in Fig. 12(b).

When corrosion time exceeds optimal parameters, the fracture morphology illustrated in Fig. 11(c) exhibits large-scale void-type filler infiltration zones within the silica matrix. Combined with Fig. 2(e), these features confirm that the filler penetrated pre-existing voids in the composite. The severe thermal mismatch between the AgCuInTi and SiO_{2f}/SiO₂, coupled with extensive infiltration, induces extremely high residual stresses in the joint, and cracks tend to propagate through these infiltration pathways. Similarly, under excessive brazing temperatures (Fig. 11(d)), fiber exposure on the fracture surface with substantial filler residue between SiO₂ fibers is observed. This phenomenon originates from excessive braze infiltration depth, which induces localized thermal mismatch stresses at the silica fiber/filler interfaces. These concentrated stresses drive inter-fiber crack propagation, ultimately degrading joint strength.

4. Conclusions

In this study, an innovative strategy for surface activation of SiO_{2f}/SiO₂ composites using thermal corrosion is proposed. The optimized corrosion time is 10 min, which not only improves the wettability of the SiO_{2f}/SiO₂ composites but also enhances the mechanical strength of the sapphire-SiO_{2f}/SiO₂ joints. The specific conclusions are as follows:

- (1) NaOH thermal corrosion can increase surface roughness and activity of the SiO_{2f}/SiO₂ composites, effectively improving the wettability of SiO_{2f}/SiO₂, reducing the contact angle from 71° to 33°, which ensures the interfacial reactions and wetting between AgCuInTi and SiO_{2f}/SiO₂. However, excessive corrosion duration induced excessive consumption of the silica matrix, accompanied by the formation of surface pores.
- (2) During the brazing process, interfacial reactions occur between the filler metal and both substrates, resulting in a typical microstructure of the brazed joint: $\alpha\text{-Al}_2\text{O}_3/\text{Ti}_2\text{O} + \text{Cu}_3\text{Ti}_3\text{O}/\text{Ag}(\text{s},\text{s}) + \text{Cu}(\text{s},\text{s})/\text{Cu}_3\text{Ti}_3\text{O} + \text{Ti}_5\text{Si}_x$ ($x = 2, 3, \text{ and } 4$) + Na₂Ti₉O₁₉/ SiO_{2f}/SiO₂.
- (3) The interfacial reaction layer thickness increases progressively with elevated brazing temperatures. When the temperature reaches 840 °C, the brittle Cu₄Ti₃ intermetallic phase forms within the braze seam. Further increasing the brazing temperature results in the development of multiple cracks within the reaction layer.
- (4) At a brazing temperature of 820 °C and corrosion time of 10 min, the joint reaches a maximum shear strength of 52 MPa. The fracture surface reveals extensive fiber pull-out traces and delamination patterns. This indicates that the crack does not propagate along a single interface, and the fracture originates from the SiO_{2f}/SiO₂ composites side, which are all typical characteristics of a high-strength fracture morphology.

CRedit authorship contribution statement

Zhenyu Ye: Writing – review & editing, Writing – original draft, Supervision, Investigation, Formal analysis, Data curation. **Yong Xia:** Data curation. **Meng Liu:** Data curation, Conceptualization. **Zijia Wang:** Writing – review & editing, Conceptualization. **Mengting Yang:** Data curation, Conceptualization. **Jiping Chen:** Conceptualization. **Peixin**

Li: Writing – review & editing, Methodology. **Jinchun Tu:** Resources, Project administration. **Jian Cao:** Resources, Project administration. **Yaotian Yan:** Resources, Project administration. **Junlei Qi:** Resources, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was supported by the National Natural Science Foundation of China (Nos. [52175303](#) and [52505362](#)), the National Science Fund for Distinguished Young Scholars (No. [52125502](#)), the Postdoctoral Fellowship Program of CPSF (No. [GZB20240949](#)), the National Natural Science Foundation of China (No. [524B200166](#)), the Key Research and Development Project of Heilongjiang Province (No. [2024ZXDXA19](#)), the Hainan Province Science and Technology Special Fund (No. [ZDYF2024SHFZ082](#)), the State Key Laboratory of Precision Welding & Joining of Materials and Structures (Nos. [24-R-01](#) and [24-Z-07](#)), the Natural Science Foundation of Heilongjiang Province (No. [LH2024E031](#)), the Heilongjiang Postdoctoral Fund (No. [LBH-Z24174](#)). Special thanks to HIT Center of Analysis Measurement and Computing, Dr. Zheng Zhen and Dr. Guo Shu for the electron microscope test analysis.

Appendix. Supplementary materials

 [Download: Download Word document \(2MB\)](#)

References

- [1] C. Zou, B. Li, X. Meng, K. Liu, X. Yang, D. Li, S. Wang
Corros. Sci., 139 (2018), pp. 243-254

[View PDF](#)[View article](#)[View in Scopus ↗](#)[Google Scholar ↗](#)

- [2] L. Yao, Z. Liu, Q. Song, B. Wang, Y. Cai, J. Zhao
J. Mater. Res. Technol., 25 (2023), pp. 5917-5937

[View PDF](#)[View article](#)[View in Scopus ↗](#)[Google Scholar ↗](#)

- [3] C. Yang, Y. Pan, W. Li, F. Xu
Ceram. Int., 51 (2025), pp. 8329-8338

[View PDF](#)[View article](#)[View in Scopus ↗](#)[Google Scholar ↗](#)

- [4] D. Li, B. Zhang, L. Zhang, Q. Chang, Z. Sun
J. Eur. Ceram. Soc., 45 (2025), Article 117181

[View PDF](#)[View article](#)[View in Scopus ↗](#)[Google Scholar ↗](#)

- [5] C. Li, Y. Piao, F. Zhang, Y. Zhang, Y. Hu, Y. Wang
Int. J. Extrem. Manuf., 5 (2023), pp. 15101-15252

[Google Scholar ↗](#)

- [6] Q. Wei, J. Lin, H. Shi, G. Tang, W. Chai, L. Qin
ACS Appl. Mater. Interfaces, 10 (2018), pp. 7693-7696

[Crossref ↗](#)[View in Scopus ↗](#)[Google Scholar ↗](#)

- [7] W. Li, T. Liang, W. Liu, C. Lei, Y. Hong, Y. Li, Z. Li, J. Xiong
Appl. Surf. Sci., 494 (2019), pp. 566-574

[View PDF](#)[View article](#)[View in Scopus ↗](#)[Google Scholar ↗](#)

- [8] X. Liu, Y. Zhang, Q. Li, J. Zheng, Y. Lu, S. Juodkazis, Q. Chen, H. Sun
Photonix, 3 (2022), p. 1

[View PDF](#)[View article](#)[Google Scholar ↗](#)

- [9] P. Wang, W. Liu, L. Gu, H. Ran, X. Song, Z. Pan, H. Chen, W. Li
Trans. Mater. Res., 1 (2025), Article 100012

[View PDF](#)[View article](#)[Google Scholar ↗](#)

- [10] T. Luo, Z. Chao, L. Jiang, S. Chen, S. Li, Y. Meng, H. Han, S. Du, B. Han, R. Zhang, M. Liu, G. Chen
J. Sci. Adv. Mater. Dev., 10 (2025), Article 100941
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [11] P. Li, Z. Wang, Y. Chen, H.A. Butt, Z. Zhang, Z. Huang, T. Zhang, L. Qiao, Y. Zhong, D. Krasnikov, A. Nasibulin, J. Cao, Y. Yan, J. Qi
Carbon, 242 (2025), Article 120416
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [12] J. Yang, W. Zhao, P. Lin, Q. Zhang, X. Zhang, W. Liu, T. Lin, H. Peng, Y. Zhuang
Mater. Des., 224 (2022), Article 111377
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [13] L. Li, Y. Ren, S. Chang, M. Li, S. Lin, M. Wang
Mater. Sci. Eng. A, 885 (2023), Article 145540
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [14] Y. Yan, P. Li, Z. Zhang, Y. Wang, J. Zhang, L. Qiao, J. Cao, J. Qi
Compos. Pt. B-Eng., 282 (2024), Article 111557
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [15] A. Pramanik, A.K. Basak, Y. Dong, P.K. Sarker, M.S. Uddin, G. Littlefair, A.R. Dixit, S. Chattopadhyaya
Compos. Pt. A-Appl. Sci. Manuf., 101 (2017), pp. 1-29
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [16] J. Fang, L. Sun, Q. Qi, J. Zhang, C. Liu
J. Eur. Ceram. Soc., 39 (2019), pp. 4545-4553
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [17] Q. Ma, Z.R. Li, S.L. Chen, D.L. Luo, Z. Zhou, J.H. Lin, J.L. Qi
J.C. Feng, Mater. Lett., 182 (2016), pp. 159-162
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [18] B. Chen, W. Zou, W. Li, S. Wu, H. Xiong, X. Wu
J. Mater. Sci. Technol., 50 (2020), pp. 13-20
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [19] J. Yang, X. Zhang, G. Ma, P. Lin, Y. Xu, T. Lin, P. He, W. Long, J. Li
J. Eur. Ceram. Soc., 41 (2021), pp. 2312-2322
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [20] X. Bi, J. Li, P. Li, H. Dong
Compos. Pt. B-Eng., 302 (2025), Article 112560
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [21] Y. Zhou, D. Liu, H. Niu, X. Song, X. Yang, J. Feng
Mater. Des., 93 (2016), pp. 347-356
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [22] Y. Wang, Z. Yong, K. Sun, Q. Wen, S. Niu, Z. Yang
Mater. Charact., 221 (2025), Article 114742
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [23] P. Wang, W. Liu, J. Li, W. Shi, Z. Liu, S. Zhao, X. Nai, H. Chen, Q. Wang, W. Li
Appl. Surf. Sci., 652 (2024), Article 159257
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [24] J. Ba, X. Ji, B. Wang, P. Li, J. Lin, J. Qi, J. Cao
Compos. Pt. B-Eng., 218 (2021), Article 108942
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [25] Z. Sun, L.X. Zhang, Q. Chang, Z.H. Zhang, T.D. Hao, J.C. Feng
J. Mater. Process. Technol., 255 (2018), pp. 8-16
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [26] O. Dezellus, F. Hodaj, N. Eustathopoulos
Acta Mater., 50 (2002), pp. 4741-4753

[View PDF](#)[View article](#)[View in Scopus ↗](#)[Google Scholar ↗](#)

- [27] V.L. Niranjani, B.S.S.C. Rao, V. Singh, S.V. Kamat
Mater. Sci. Eng. A, 529 (2011), pp. 257-264

[View PDF](#)[View article](#)[View in Scopus ↗](#)[Google Scholar ↗](#)

- [28] R. Asthana
Metall. Mater. Trans. A, 25 (1994), pp. 225-230

[View in Scopus ↗](#) [Google Scholar ↗](#)

- [29] L. Zhang, Z. Sun, Q. Chang, J. Shi, J. Feng
Ceram. Int., 45 (2019), pp. 1698-1709

[View PDF](#)[View article](#)[View in Scopus ↗](#)[Google Scholar ↗](#)

- [30] Z. Sun, Y. Cao, L. Zhang, J. Feng, J. Li, J. Eur
Ceram, 40 (2020), pp. 1488-1495

[View PDF](#)[View article](#)[View in Scopus ↗](#)[Google Scholar ↗](#)

- [31] Z. Sun, L. Zhang, J. Qi, Z. Zhang, C. Tian, J. Feng
Mater. Des., 88 (2015), pp. 51-57

[View PDF](#)[View article](#)[View in Scopus ↗](#)[Google Scholar ↗](#)

- [32] Z. Sun, L. Zhang, T. Hao, J. Feng
Ceram. Int., 44 (2018), pp. 15809-15816

[View PDF](#)[View article](#)[View in Scopus ↗](#)[Google Scholar ↗](#)

- [33] J. Yang, W. Zhao, P. Lin, Q. Zhang, X. Zhang, T. Lin, P. He, Y. Zhuang
J. Mater. Sci. Technol., 151 (2023), pp. 234-244

[View PDF](#)[View article](#)[Crossref ↗](#)[Google Scholar ↗](#)

- [34] L. Zhang, L. Wu, D. Liu, J. Feng, H. Liu
Mater. Sci. Eng. A, 496 (2008), pp. 393-398

[View PDF](#)[View article](#)[View in Scopus ↗](#)[Google Scholar ↗](#)

- [35] J.H. Schneibel, C.J. Rawn
Acta Mater., 52 (2004), pp. 3843-3848
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [36] J. Zhang, Y. Yan, Z. Zhang, Y. Xia, Z. Ye, B. Wang, P. Li, P. Wang, J. Lin, H.A. Butt, D.V. Krasnikov, A.G. Nasibulin, J. Tu, J. Cao, J. Qi
Ceram. Int., 50 (2024), pp. 27276-27286
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [37] L. Wu, X. Wang, X. Feng, B. Chai, Y. Mao, L. Fu
Ceram. Int., 50 (2024), pp. 50569-50586
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [38] P. Wang, W. Shi, L. Gu, H. Ran, Z. Pan, Y. Su, X. Song, J. Cao, H. Chen, W. Li
J. Mater. Sci. Technol., 244 (2025), pp. 46-59
[Google Scholar ↗](#)

[View Abstract](#)

Recommended articles

Based on reading popularity

Research article Full text access

First-principles study of the ferromagnetism of Mn-doped 3C-SiC

Sheng-An Tang, ..., Chao Zhang

Solid State Communications • Volume 321 • 2020 • Article 114039

 [View PDF](#)

Research article Full text access

Effect of the fabrication method on the wear properties of copper silicon carbide composites

Mohamed Ibrahim Abd El Aal, H.S. Kim

Tribology International • Volume 128 • 2018 • Pages 140-154



[View PDF](#)

Research article Full text access

Growth kinetics of interfacial reaction layer products between cubic boron nitride and Cu-Sn-Ti active filler metal

Yonggang Fan, Cong Wang

Journal of Materials Science & Technology • Volume 92 • 2021 • Pages 69-74



[View PDF](#)

Research article Full text access

Deformation induced precipitation of MgZn₂-type laves phase in Ti-Fe-Co alloy

L.M. Kang, ..., W.W. Zhang

Journal of Alloys and Compounds • Volume 778 • 2019 • Pages 795-802



[View PDF](#)

Short communication Full text access

Effect of the precipitated phase of MgZn₂ on surface nanocrystallization of Al-Zn-Mg alloy based on high-frequency impacting and rolling

Xiaohui Zhao, ..., Furong Chen

Materials Letters • Volume 188 • 2017 • Pages 95-98



[View PDF](#)

Research article Full text access

Influence of C addition on solid-state reaction of Ti-Si-C system

Kian Kasraee, ..., Seyed Ali Tayebifard

Materials Chemistry and Physics • Volume 217 • 2018 • Pages 379-387



[View PDF](#)

Cited by (5)

Research article Open access

Wettability and metallurgical reactions of new experimental Fe-P based and Mn-Fe-P based brazing alloys on non-alloy and low-alloy steels

Zorc M., ..., Zorc B.

Journal of Advanced Joining Processes • Volume 13 • 2026 • Article 100400[Show abstract](#) ✓

Research article

Achieving a synergistic combination of high hardness, enhanced creep resistance and damage tolerance in plasma-sprayed eutectic/amorphous Al₂O₃-YAG composite coatings through laser remelting

Yang K., ..., Gao Y.

Surface and Coatings Technology • Volume 524 • 2026 • Article 133278[Show abstract](#) ✓

Research article

Microstructure and mechanical properties of HVOF-sprayed WC-based cermet composite coating reinforced by MWCNT@MXene

Zhang Z., ..., Hong S.

Surface and Coatings Technology • Volume 521 • 2026 • Article 133071[Show abstract](#) ✓

Research article Open access

Low-temperature and low-pressure sinter-bonding for thermoelectric generator devices using Cu nanoparticle paste

Chung S., ..., Kim D.

Materials Today Communications • Volume 51 • 2026 • Article 114755[Show abstract](#) ✓

Research article Open access

Hydrogen-assisted enhancement of diffusion bonding for Zr-4 alloys with a Ti interlayer

Bai Y., ..., Li Y.

Journal of Materials Research and Technology • Volume 40 • 2026

All content on this site: Copyright © 2026 Elsevier B.V., its licensors, and contributors. All rights are reserved, including those for text and data mining, AI training, and similar technologies. For all open access content, the relevant licensing terms apply.

Mendeley Readers

4



[View details ↗](#)

© 2026 Published by Elsevier Ltd on behalf of The editorial office of Journal of Materials Science & Technology.